

How to Store Triple Power in Warehouse



Since the lithium metal is a particularly susceptible to chemical reaction, and burning is easy to extend, the lithium battery packaging and transport, if handled improperly, combustible and explosive, accidents have occurred, so the lithium battery Laid warehouse management requirements as follows:

1. Basic requirements

a) Due to the characteristics of lithium batteries, the storage environment temperature needs to meet the requirements:

T-BAT-SYS-HV-4.5/6.3:

It can be stored at -20°C to 45°C for 3 months and at 0°C to 40°C for 1 year.

T-BAT-SYS-HV-5.8:

It can be stored at -20°C to 30°C for 1 year, and at 30°C to 55°C for 6 months.

T-BAT-SYS-HV-3.0; T-BAT-SYS-HV-R2.5/3.6; T-BAT-SYS-HV-5.0;

T-BAT-SYS-HV-S2.5/3.6; T-BAT-SYS-LV-R2.5/3.6; T-BAT-SYS-HV-S50E;

T-BAT-SYS-LV D53/143/153; TSYS-HS51:

It can be stored at -20°C to 30°C for 1 year, and at 30°C to 50°C for 6 months.

b) Humidity requirements: effectively control the humidity in the warehouse to avoid extreme humidity in the warehouse for a long time (relative humidity higher than 90% or lower than 40%).

c) Lithium battery warehouses should be physically separated by brick walls, and the warehouses must be enclosed, explosion-proof or other corresponding safety electrical lighting equipment.

d) Places where flammable and explosive materials are stored should be equipped with a sufficient number of fire-fighting equipment and be in good condition.

e) Where there are lithium batteries, there must be some illegal regulations such as prohibiting smoking.

f) It is forbidden to carry out operations that may cause fire in warehouses and places where flammable and explosive materials are stored.

2. Storage requirements

a) Lithium batteries should be stored in a well-ventilated, dry and cool place. High temperature and high humidity may damage the battery performance and corrode the battery surface.

b) Battery cartons should not be stacked higher than the specified height. If too many battery cartons are stacked together, the batteries in the bottom cartons may be deformed and liquid leakage may occur.

c) Avoid storing or displaying the battery in direct sunlight or a place exposed to rain. If the battery is drenched, the insulation resistance will decrease, and self-discharge and rust may occur. A rise in temperature may damage the battery.

d) Store and display the batteries in the original packaging. After removing the packaging, the batteries will be stacked randomly, which may cause short circuit and damage to the batteries.

e) Items that are likely to cause combustion or explosion when they come into contact with each other and items with different fire-fighting methods should be stored separately.

3. Handling requirements

a) When handling materials, no matter what kind of handling tools are used, they should be handled properly to prevent materials from falling or being damaged.

b) When carrying out material handling, load, stacking, directionality and other issues should be considered.

c) When using forklifts or carts, the loading method should be heavy and light, and the material to be transported should not exceed the width and height of the passage and elevator doors, and the speed should be moderate.

d) Transporters should use qualified transport tools (forklifts, carts, etc).

e) The battery cardboard box should be handled with care. Rough handling may cause short circuit or damage to the battery, resulting in leakage, explosion or fire.

4. Processing requirements

a) In the case of not inconsistent with local regulations, lithium (primary) batteries can be used for public garbage disposal.

b) Do not disassemble the battery.

c) Except for the use of approved controllable stoves, it is forbidden to dispose of batteries by fire.

5. Warehouse requirements

a) Warehouse management personnel should check the cargo information every day. If they find that the storage location is incorrect, the accounts are inconsistent, and the quality problems are reported and dealt with in a timely manner, they should conduct a fire prevention inspection and cut off the power supply at the end of work or after get off work.

b) Clean up the warehouse area every day, clean up dirt and debris on the ground in time, and arrange the materials in the warehouse into the designated area to meet the requirements of neat, tidy, clean, hygienic and reasonable placement.

c) Keep safe passages in the warehouse unblocked to prevent accumulations and ensure the safety of personnel.

d) The planning area in the warehouse must be clearly marked, and the material storage area must be classified and stored in different districts, and clearly marked.

6. Delivery requirements

- a) The battery packs are shipped on a first-in-first-out basis.

- b) Use the same batch group together, and try not to use different batches in combination(it will affect the service life of the overall system and the after-sales problem of battery imbalance).

- c) For the shipment of different batches of batteries, if the time interval between batches and batches is greater than or equal to 3 months or more, the batteries in this group need to be fully charged before they can be shipped(In case the battery capacity is uneven when the battery reaches the user end, please refer to the charging manual for details).

- d) Counting the storage time, the battery packs that have accumulated up to 6 months after storage are still in the warehouse (not required for the main control), and battery charging operations are required (For specific operation methods, please refer to the charging operation manual).

- e) Count the delivery time and inform customers that they need to be stored in accordance with our requirements. Similarly, if the storage time in their warehouse reaches 6 months, they need to be charged (Time starts from our delivery time).

7. Emergency treatment method

Lithium batteries may leak, rust and expand when stored for a long time; if they are not handled properly, they may heat up, burn or explode. The relevant treatment methods are as follows:

a) Rust treatment method: usually see cylindrical lithium battery (polymer lithium battery does not have this phenomenon), the initial, slight rust will not affect the performance of the lithium battery, and it can be used normally. If the rust is serious (such as the part of the cap), it will affect the sealing performance of the battery and leak, and must be scrapped.

b) Treatment of leakage or swelling: leakage refers to the leakage of electrolyte in the battery, which usually has a pungent odor. The strong corrosiveness of the electrolyte will cause damage to the battery protection board components, such as polymer Lithium batteries will swell. Leaking and swollen batteries must be sorted out and discarded.

c) Under normal environmental conditions of temperature and humidity, the battery will not produce mildew or discoloration. If leakage occurs, such undesirable phenomena will flourish.

d) Heat treatment method: it will generate heat during charging and discharging (under use), but the temperature is usually below 60 degrees.

The temperature of the battery can reach hundreds or hundreds of hundreds in the state of internal or external short circuit. At this time, the battery must be isolated and placed in the sand. Be careful not to directly touch the battery with your hands, otherwise it will be burned. After the battery temperature drops to normal temperature, discard it.

e) Combustion or explosion treatment method: If combustion or explosion occurs, it is very dangerous, and personnel must stay away; battery burning will not produce open flames, usually due to high levels of ignition of nearby flammable products or packaging outer boxes, take protective measures Under the premise, such as the fire extinguishing method of a single or very few batteries burning or exploding; use sand to directly cover the burning or exploding battery. If it is a large area of burning, use a dry powder fire extinguisher to extinguish the fire. It is strictly forbidden to use water to extinguish the fire, because water will cause the battery to short circuit.